A Barvinok–Pataki Theorem for Archimedean Order Unit Spaces

Timo Ziegler,[1] Tobias J. Osborne,[1] Martin Plavala,[2]
René Schwonnek,[1] Lennart Binkowski,[1] Lukas Hantzko,[2]
Henrik Wilming,[1] Alexander Stottmeister[3]

[1]AG Osborne, Institut für Theoretische Physik, Leibniz Universität Hannover
[2]AG Raussendorf, Institut für Theoretische Physik, Leibniz Universität Hannover
[3]Quantum Information Group, Institut für Theoretische Physik, Leibniz Universität Hannover

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Abstract

We prove a generalisation of the Barvinok–Pataki rank bound to Archimedean order unit spaces equipped with a pointed cone. The classical result bounds the rank of extreme points of spectrahedra (feasible sets of semidefinite programs); our generalisation bounds the face dimension of feasible points under affine constraints in an arbitrary pointed cone, including the infinite-dimensional case. The face-dimension bound $d_F \leq m$ uses only pointedness of the cone and rank–nullity; finite-dimensionality of d_F in infinite dimensions follows from a result of Weis–Shirokov that requires only a real vector space and a convex set. When specialised to C*-algebras via the Alfsen–Shultz face–commutant correspondence, this recovers the operator-algebraic Barvinok–Pataki bound $\sum n_j^2 \leq m + 1$.

1 Preliminaries

Definition 1 (Ordered vector space with cone)**.** Let V be a real vector space. A *cone* is a subset $V^+ \subseteq V$ that is closed under addition and non-negative scalar multiplication. The pair (V, V^+) is an *ordered vector space* with partial order $x \leq y$ iff $y - x \in V^+$.

Definition 2 (Pointed cone)**.** A cone V^+ is *pointed* (or *proper*) if $V^+ \cap (-V^+) = \{0\}$, i.e. the only element v with both $v \geq 0$ and $-v \geq 0$ is $v = 0$.

Definition 3 (Order unit and Archimedean property)**.** An element $e \in V^+$ is an *order unit* if for every $v \in V$ there exists $n \in \mathbb{N}$ such that $ne - v \in V^+$ (i.e. $v \leq ne$). The cone V^+ is *Archimedean* if, whenever $ne + v \in V^+$ for all $n \in \mathbb{N}$, it follows that $v \in V^+$. The triple (V, V^+, e) is called an *Archimedean order unit space* (AOU space) if V^+ is a pointed Archimedean cone and e is an order unit.

Definition 4 (Face of a cone)**.** Let V^+ be a cone in V and $x \in V^+$. The *face of* V^+ *generated by* x is

$$\mathrm{face}_{V^+}(x) \;=\; \left\{\, w \in V^+ \;\middle|\; \exists\, \lambda > 0 \colon x - \lambda w \in V^+ \,\right\}.$$

Equivalently, $\mathrm{face}_{V^+}(x)$ consists of all elements of V^+ that can be "majorised" by a positive multiple of x. This is the smallest face of V^+ containing x in its relative interior.

Remark 5. In a pointed cone, $\mathrm{face}_{V^+}(x) = \{0\}$ if and only if $x = 0$. For $x \neq 0$, the face is at least one-dimensional (it contains the ray $\{\lambda x : \lambda \geq 0\}$). For the PSD cone PSD_n, the face generated by a rank-r matrix X is isomorphic to PSD_r, with $\dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$ (real symmetric case) or r^2 (complex Hermitian case).

2 The Classical Barvinok–Pataki Bound

We first recall the classical proof for PSD matrices, as it motivates each step of the generalisation.

Theorem 6 (Barvinok–Pataki, PSD case)**.** *Let $A_1, \dots, A_m \in \mathrm{Sym}_n(\mathbb{R})$ and $\beta_1, \dots, \beta_m \in \mathbb{R}$. Consider the spectrahedron*

$$C = \{\, X \in \mathrm{PSD}_n \mid \langle A_j, X\rangle = \beta_j,\ j = 1, \dots, m \,\}.$$

If $X \in C$ is an extreme point of C, then $\frac{r(r+1)}{2} \le m$ where $r = \mathrm{rank}(X)$.

Proof. Write $X = VBV^\top$ with $V \in \mathbb{R}^{n\times r}$, $B \in \mathrm{PSD}_r$ strictly positive definite. Define the compressed constraints $\hat{A}_j = V^\top A_j V \in \mathrm{Sym}_r$ and the linear map

$$\varphi\colon \mathrm{Sym}_r \to \mathbb{R}^m, \qquad [\varphi(D)]_j = \langle \hat{A}_j, D\rangle.$$

By rank–nullity, $\dim(\ker\varphi) \ge \frac{r(r+1)}{2} - m$. If $\ker\varphi \ne \{0\}$, choose $D \ne 0$ in $\ker\varphi$. Then $B(t) = B + tD$ satisfies all constraints for every t.

Since $B \succ 0$ and D is a non-zero symmetric matrix with at least one negative eigenvalue (replacing D by $-D$ if necessary), the map $t \mapsto \lambda_{\min}(B + tD)$ is continuous, positive at $t = 0$, and decreasing for $t > 0$. Let $t_0 > 0$ be the first value where $\lambda_{\min}(B + t_0 D) = 0$. Then $B(t_0) \succeq 0$ with $\mathrm{rank}(B(t_0)) < r$, and $VB(t_0)V^\top \in C$.

This contradicts X being an extreme point of C (since X is a non-trivial convex combination of $VB(t_0)V^\top$ and $VB(-t_0')V^\top$ for suitably chosen t_0'), unless $\ker\varphi = \{0\}$, which forces $\frac{r(r+1)}{2} \le m$. □

3 Generalisation to Pointed Cones

The key observation is that the proof above uses only three ingredients:

1. **Rank–nullity** for the constraint map restricted to the face;
2. **Relative interior perturbation**: any kernel direction $D \in \mathrm{span}(F)$ can perturb x in both directions while remaining in the face (since x lies in the relative interior of F);
3. **Finite-dimensionality** of the face, to apply rank–nullity with finite dimensions.

Pointedness of the cone is used implicitly: it ensures that the face structure of V^+ is non-trivial (every face is a proper sub-cone).

Theorem 7 (Barvinok–Pataki for pointed cones)**.** *Let V be a real vector space, $V^+ \subseteq V$ a pointed cone, and $\alpha_1, \dots, \alpha_m \in V^*$ linear functionals. Define*

$$C = \{\, x \in V^+ \mid \alpha_j(x) = b_j,\ j = 1, \dots, m \,\}$$

for given $b_1, \dots, b_m \in \mathbb{R}$. Let $x \in C$ and let $F = \mathrm{face}_{V^+}(x)$ be the face of V^+ generated by x. Assume that $d_F := \dim(\mathrm{span}(F))$ is finite. Define the restricted constraint map

$$\varphi_F\colon \mathrm{span}(F) \to \mathbb{R}^m, \qquad [\varphi_F(v)]_j = \alpha_j(v).$$

If x is an extreme point of C, then

$$\boxed{d_F \le m.}$$

Equivalently, $\dim(F) \le m$ (where $\dim(F) = \dim(\mathrm{aff}(F)) = d_F$, since $0 \in F$ for every face of a cone).

Remark 8 (Real symmetric vs. complex Hermitian face dimensions)**.** When specialised to PSD matrices, the face dimension d_F depends on whether one works over $\mathbb{R}$ or $\mathbb{C}$. For a rank-r matrix X:

- **Real symmetric** ($X \in \mathrm{Sym}_r(\mathbb{R})$): $d_F = \dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$, giving $\frac{r(r+1)}{2} \leq m$.
- **Complex Hermitian** ($X \in \mathrm{Herm}_r(\mathbb{C})$): $d_F = \dim_{\mathbb{R}}(\mathrm{face}(X)) = r^2$, giving $r^2 \leq m$.

All numerical SDPs in this paper use **real symmetric** PSD variables (with BP formula $r(r+1)/2 \leq m$), since the JuMP/Mosek interface parameterises PSD variables as real symmetric matrices. This restricts numerics to channels with real Kraus operators; the analytic results in Sections 13 and 12 use the complex Hermitian formula $r^2 \leq m$ where appropriate.

Proof. By rank–nullity applied to φ_F,

$$d_F \;=\; \dim(\ker \varphi_F) + \dim(\mathrm{im}\, \varphi_F) \;\leq\; \dim(\ker \varphi_F) + m.$$

It suffices to show that $\ker \varphi_F = \{0\}$ when x is extreme in C. Suppose for contradiction that there exists $D \in \ker \varphi_F$, $D \neq 0$.

Step 1: The perturbation preserves feasibility. Since $D \in \mathrm{span}(F) \cap \ker \varphi_F$, we have $\alpha_j(D) = 0$ for all j. Therefore, for any $t \in \mathbb{R}$:

$$\alpha_j(x + tD) = \alpha_j(x) + t\,\alpha_j(D) = b_j.$$

So $x + tD$ satisfies all affine constraints whenever it lies in V^+.

Step 2: The point x lies in the algebraic relative interior of F. The face $F = \mathrm{face}_{V^+}(x)$ is by definition the smallest face of V^+ containing x in its relative interior. In particular, for every $D \in \mathrm{span}(F)$ there exists $\varepsilon > 0$ such that $x + tD \in F \subseteq V^+$ for all $|t| < \varepsilon$.

To see this explicitly: write $D = w_1 - w_2$ with $w_1, w_2 \in F$. By definition of F, there exist $\lambda_1, \lambda_2 > 0$ with $x - \lambda_i w_i \in V^+$. For $0 < t < \min(\lambda_1, \lambda_2)$,

$$x + tD = \underbrace{(x - tw_2)}_{\in V^+} + \underbrace{tw_1}_{\in V^+} \;\in\; V^+,$$

where $x - tw_2 = (1 - t/\lambda_2)\, x + (t/\lambda_2)(x - \lambda_2 w_2) \in V^+$ by convexity. The same argument applies to $x - tD$.

Step 3: Contradiction with extremality. Combining Steps 1 and 2: for sufficiently small $\varepsilon > 0$, both $x^+ := x + \varepsilon D$ and $x^- := x - \varepsilon D$ lie in $F \subseteq V^+$ and satisfy all constraints, hence $x^+, x^- \in C$. Since $D \neq 0$, they are distinct ($x^+ - x^- = 2\varepsilon D \neq 0$). Then

$$x \;=\; \tfrac{1}{2}\, x^+ \;+\; \tfrac{1}{2}\, x^-$$

is a non-trivial convex combination of two distinct elements of C, contradicting the extremality of x.

Therefore $\ker \varphi_F = \{0\}$, and we conclude $d_F \leq m$. □

4 Application to C*-Algebras and AOU Spaces

Corollary 9 (Barvinok–Pataki for AOU spaces)**.** *Let (V, V^+, e) be an Archimedean order unit space with $\dim(V) < \infty$. Let $\alpha_1, \ldots, \alpha_m \in V^*$ and consider the state-like set*

$$C = \left\{\, x \in V^+ \;\middle|\; \alpha_j(x) = b_j,\ j = 1, \ldots, m \,\right\}.$$

For any extreme point x of C, the face of V^+ generated by x satisfies $d_F = \dim(\mathrm{span}(\mathrm{face}_{V^+}(x))) \leq m$.

Proof. By definition, an AOU space has a pointed cone. The Archimedean property implies V^+ is closed in the order-unit norm. In finite dimensions, every face is finite-dimensional. Apply Theorem 7. □

Corollary 10 (Barvinok–Pataki for C*-algebras)**.** *Let A be a unital C*-algebra and $a_1, \ldots, a_m \in A_{\mathrm{sa}}$. Consider the affinely constrained state space*

$$C = \{\, \varphi \in S(A) \mid \varphi(a_j) = \beta_j,\ j = 1, \ldots, m \,\},$$

where $S(A) = \{\varphi \in A^* : \varphi \geq 0,\ \varphi(\mathbf{1}) = 1\}$. If φ is an extreme point of C, then

$$\dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} \ \leq\ m + 1,$$

where π_φ is the GNS representation of φ. Equivalently, if $\pi_\varphi(A)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ (Artin–Wedderburn), then $\sum_j n_j^2 \leq m + 1$.

Proof. The state space $S(A)$ is a base for the cone A_+^* (positive functionals), cut out by the normalisation constraint $\varphi(\mathbf{1}) = 1$. The total number of affine constraints on A_+^* is $m + 1$ (the m given constraints plus normalisation).

By the Alfsen–Shultz face–commutant correspondence, the face of $S(A)$ generated by φ is affinely isomorphic to the state space $S(\pi_\varphi(A)')$ of the commutant. In particular,

$$\dim\big(\mathrm{face}_{S(A)}(\varphi)\big) = \dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} - 1.$$

The cone A_+^* is pointed (if φ and $-\varphi$ are both positive, then $\varphi(a^*a) = 0$ for all a, so $\varphi = 0$) and weak-$*$ closed. When A is infinite-dimensional, the finiteness of d_F follows from Corollary 12 below (applied with $\ell = m + 1$). Theorem 7 then gives $\dim(\mathrm{span}(\mathrm{face}(\varphi))) \leq m + 1$, hence $\dim_{\mathbb{R}}(\pi_\varphi(A)')_{\mathrm{sa}} \leq m + 1$. □

5 The Infinite-Dimensional Case and the Role of d_F

Theorem 7 assumes that $d_F = \dim(\mathrm{span}(F))$ is finite. In infinite dimensions this is a non-trivial hypothesis: the face of a cone generated by a point can be infinite-dimensional. We now show that this hypothesis is *automatically satisfied* whenever the number of constraints is finite, using a result of Weis–Shirokov that is purely convex-geometric.

Theorem 11 (Weis–Shirokov [4, Theorem 3.3])**.** *Let V be a real vector space, $K \subseteq V$ a convex set, and $f\colon K \to \mathbb{R} \cup \{+\infty\}$ a generalised affine map. Let C denote a sublevel set $\{x \in K : f(x) \leq \alpha\}$ or a level set $\{x \in K : f(x) = \alpha\}$, and let $x \in C$. If the face $F_C(x)$ of C generated by x has dimension $m \in \mathbb{N}_0$, then the face $F_K(x)$ of K generated by x has dimension m or $m + 1$.*

The multi-constraint version follows by induction (Corollary 5 of [4]): if C is the intersection of ℓ sublevel and/or level sets of generalised affine maps, and $F_C(x)$ has dimension m, then $\dim(F_K(x)) \in \{m, m + 1, \ldots, m + \ell\}$.

Corollary 12 (Finiteness of d_F)**.** *Let K be a convex set in a real vector space V, and let C be defined by ℓ generalised affine constraints on K. If x is an extreme point of C, then*

$$\dim\big(F_K(x)\big) \ \leq\ \ell.$$

In particular, the face of K generated by any extreme point of C is finite-dimensional, regardless of $\dim(V)$.

Proof. Since x is extreme in C, the face $F_C(x) = \{x\}$ has dimension 0. By the multi-constraint version of Theorem 11, $\dim(F_K(x)) \in \{0, 1, \ldots, \ell\}$. □

Remark 13 (No operator-algebraic assumptions needed)**.** Theorem 11 and Corollary 12 require only a real vector space V and a convex subset $K \subseteq V$. In particular:

- No cone structure, pointedness, or Archimedean property is needed to establish that d_F is finite.
- No C*-algebraic or operator-algebraic notions are needed.
- The operator-algebraic machinery (GNS representation, face–commutant correspondence, Artin–Wedderburn decomposition) is used only in a second step: to *interpret* the finite-dimensional face algebraically as a commutant and extract the multiplicity bound $\sum n_j^2 \leq m+1$.

This clarifies the logical structure of the infinite-dimensional case: Weis–Shirokov provides the purely geometric bound on d_F, and the Alfsen–Shultz correspondence translates it into algebra.

Combining Corollary 12 with Theorem 7, we obtain the following infinite-dimensional version of the main theorem that does not require an a priori finiteness assumption on d_F.

Corollary 14 (Barvinok–Pataki for infinite-dimensional pointed cones)**.** *Let V be a (possibly infinite-dimensional) real vector space, $V^+ \subseteq V$ a pointed closed cone, and $\alpha_1, \ldots, \alpha_m \in V^*$ linear functionals. Define $C = \{x \in V^+ : \alpha_j(x) = b_j\}$. If x is an extreme point of C, then*

$$d_F = \dim\big(\mathrm{span}(\mathrm{face}_{V^+}(x))\big) \leq m.$$

Proof. By Corollary 12 (with $K = V^+$ viewed as a convex set and $\ell = m$), d_F is finite. Theorem 7 then gives $d_F \leq m$. □

6 Special Cases Recovered

1. **Classical Barvinok–Pataki** ($A = M_n(\mathbb{C})$). The unique irreducible representation is the identity, so $\pi_\varphi(A)' \cong M_r(\mathbb{C})$ where $r = \mathrm{rank}(\rho_\varphi)$. Hence $r^2 \leq m+1$, or $r^2 \leq m$ when one of the constraints is the trace normalisation already built into the spectrahedron.
2. **Carathéodory's theorem** ($A = C(X)$, abelian). All irreducible representations are one-dimensional, so the commutant decomposes as $\bigoplus_{j=1}^k \mathbb{C}$. The bound gives $k \leq m+1$: extreme measures under m moment constraints are supported on at most $m+1$ points.
3. **Block-diagonal algebras** ($A = \bigoplus_i M_{d_i}(\mathbb{C})$). The bound $\sum_i r_i^2 \leq m+1$ simultaneously constrains ranks within blocks *and* the number of active blocks.
4. **Infinite-dimensional** ($A = B(\mathcal{H})$). For normal states (density operators) under m constraints, apply Corollary 14 to the pointed closed cone of positive trace-class operators: $r^2 \leq m+1$. In particular, $m \leq 2$ constraints imply all extreme points are pure states.

7 Physical Example: Hydrogen Atom in a Magnetic Field

We illustrate the theorem with a concrete physical system: the hydrogen atom in an external magnetic field, subject to angular momentum constraints. This example demonstrates how the Barvinok–Pataki bound determines when pure-state variational methods suffice and when mixed states are unavoidable.

7.1 Hilbert space and quantum numbers

The non-relativistic hydrogen atom has Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ (spatial wave function tensored with spin-$\frac{1}{2}$). The Coulomb Hamiltonian

$$H_0 = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0\, r}$$

has eigenvalues $E_n = -13.6\ \mathrm{eV}/n^2$ for $n = 1, 2, 3, \ldots$, with simultaneous eigenstates $|n, l, m_l, m_s\rangle$ labelled by:

Quantum number	**Range**	**Observable**
n (principal)	$1, 2, 3, \ldots$	H_0 (energy)
l (orbital ang. mom.)	$0, 1, \ldots, n-1$	$\mathbf{L}^2 = l(l+1)\hbar^2$
m_l (magnetic)	$-l, -l+1, \ldots, l$	$L_z = m_l\hbar$
m_s (spin)	$\pm\frac{1}{2}$	$S_z = m_s\hbar$

The degeneracy of the n-th energy shell is $g_n = 2n^2$ (including spin). The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with $J_z = L_z + S_z$ having eigenvalue $m_j\hbar$ where $m_j = m_l + m_s$.

7.2 Zeeman Hamiltonian

In a uniform magnetic field $\mathbf{B} = B\hat{z}$, the interaction Hamiltonian is

$$H_Z = \frac{\mu_B B}{\hbar}\big(L_z + g_s\, S_z\big),$$

where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton and $g_s \approx 2$ is the electron spin g-factor. On the standard basis, $H_Z|n, l, m_l, m_s\rangle = \mu_B B(m_l + 2m_s)|n, l, m_l, m_s\rangle$. The full Hamiltonian is $H = H_0 + H_Z$.

7.3 The constrained optimisation problem

We seek the state of minimum energy subject to a constraint on the total angular momentum projection:

$$\min_{\rho}\ \mathrm{Tr}(H\rho) \qquad \text{subject to} \qquad \rho \geq 0,\ \ \mathrm{Tr}(\rho) = 1,\ \ \mathrm{Tr}(J_z\,\rho) = M\hbar. \tag{1}$$

This is a convex optimisation (linear objective over a convex constraint set). The minimum is attained at an extreme point of the feasible set

$$C = \big\{\rho \in \mathfrak{S}(\mathcal{H}) : \mathrm{Tr}(J_z\,\rho) = M\hbar\big\},$$

where $\mathfrak{S}(\mathcal{H})$ is the set of density operators on $\mathcal{H}$.

7.4 Applying the Barvinok–Pataki bound

Full Hilbert space ($A = B(\mathcal{H})$, infinite-dimensional). The constraint $\mathrm{Tr}(J_z\,\rho) = M\hbar$ constitutes $m = 1$ affine constraint on the state space $\mathfrak{S}(\mathcal{H})$. By Corollary 14 applied to the pointed closed cone of positive trace-class operators (with $m + 1$ constraints including trace normalisation),

$$r^2 \ \leq\ m + 1 = 2, \qquad \text{hence} \qquad r = 1.$$

The energy minimiser is a pure state.

More generally, with m constraints:

m	**Constraints**	**Rank bound**	**Conclusion**
1	$\mathrm{Tr}(J_z\,\rho) = M\hbar$	$r = 1$	Pure state
2	$+\ \mathrm{Tr}(\mathbf{J}^2\,\rho) = j(j+1)\hbar^2$	$r = 1$	Pure state
3	$+\ \mathrm{Tr}(\mathbf{L}^2\,\rho) = l(l+1)\hbar^2$	$r \le 2$	Rank ≤ 2
4	$+\ \mathrm{Tr}(H_0\,\rho) = E$	$r \le 2$	Rank ≤ 2

The transition from pure to potentially mixed occurs at $m = 3$: this is a direct consequence of the face dimension gap for $B(\mathcal{H})$, which has $\{1, 2\} \subseteq G(B(\mathcal{H}))$ (the face dimension spectrum is $\{0, 3, 8, 15, \ldots\}$).

Restriction to the $n = 2$ shell. For a concrete finite-dimensional calculation, restrict to the $n = 2$ shell. This has $g_2 = 2 \cdot 4 = 8$ states, so $\mathcal{H}_{n=2} \cong \mathbb{C}^8$ and $A = M_8(\mathbb{C})$:

l	m_l	m_s	$m_j = m_l + m_s$	$m_l + 2m_s$	**Subshell**
0	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2s$
0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2s$
1	-1	$+\frac{1}{2}$	$-\frac{1}{2}$	0	$2p$
1	-1	$-\frac{1}{2}$	$-\frac{3}{2}$	-2	$2p$
1	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2p$
1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2p$
1	1	$+\frac{1}{2}$	$+\frac{3}{2}$	2	$2p$
1	1	$-\frac{1}{2}$	$+\frac{1}{2}$	0	$2p$

The Coulomb Hamiltonian is $H_0 = E_2\, I_8$ (degenerate on the shell). Including spin–orbit coupling (fine structure), $H_{\mathrm{fs}} = \xi_{21}\, \mathbf{L} \cdot \mathbf{S}$ splits the shell into multiplets: $2s_{1/2}$ ($l = 0$, $j = \frac{1}{2}$, 2 states), $2p_{1/2}$ ($l = 1$, $j = \frac{1}{2}$, 2 states), and $2p_{3/2}$ ($l = 1$, $j = \frac{3}{2}$, 4 states). The Zeeman term $H_Z = \mu_B B(m_l + 2m_s)$ is diagonal in the $|l, m_l, m_s\rangle$ basis.

The optimisation (1) restricted to this shell is an SDP on $M_8(\mathbb{C})$. Normalisation $\mathrm{Tr}(\rho) = 1$ is built into the state space. The Barvinok–Pataki bound gives:

m	**Constraints (on the $n = 2$ shell)**	$r^2 \le m + 1$	**Naïve bound**
1	$\mathrm{Tr}(J_z\,\rho) = M\hbar$	$r = 1$	$r \le 8$
2	$+\ \mathrm{Tr}(\mathbf{J}^2\,\rho) = j(j+1)\hbar^2$	$r = 1$	$r \le 8$
3	$+\ \mathrm{Tr}(\mathbf{L}^2\,\rho) = l(l+1)\hbar^2$	$r \le 2$	$r \le 8$
5	$+\ \mathrm{Tr}(S_z\,\rho) = s\hbar,\ \mathrm{Tr}(L_z\,\rho) = \mu\hbar$	$r \le 2$	$r \le 8$

7.5 Physical interpretation

1. **Automatic purity ($m \le 2$).** Fixing the total angular momentum projection M (and optionally j) is enough to guarantee that the energy minimiser is a pure state. This means pure-state variational methods (e.g. Rayleigh–Ritz, DMRG for the ground state) are *exact* for this problem — there is no advantage to working with density matrices.

2. **Low-rank mixed states ($m = 3$).** Adding a third constraint (e.g. fixing l) opens the door to rank-2 optimisers: the energy minimiser may be a *statistical mixture* of two pure states. But the rank is at most 2, not 8; thus only a two-dimensional search is needed, rather than optimisation over all 8×8 density matrices.

   This bound is **tight**: when the constraint values lie in the interior of the joint numerical range — i.e. values not achievable by any single pure state — the optimizer is genuinely mixed. For example, on the $n = 2$ shell with $m = 3$ constraints $\langle J_z\rangle = 0$, $\langle \mathbf{J}^2\rangle = 2$, $\langle \mathbf{L}^2\rangle = 1$ (values strictly between the eigenvalues of $\mathbf{J}^2$), the SDP optimizer has $\mathrm{rank}(\rho^*) = 2$ with eigenvalues $(\lambda_1, \lambda_2) \approx (0.569, 0.431)$. See Table 1 and Figure 1.

3. **Practical consequence.** For a numerical SDP solving (1), interior-point methods converge to the analytic centre (maximal rank), but the theorem guarantees that facial reduction or rank minimisation will find a solution of rank $\leq \lfloor\sqrt{m+1}\rfloor$. This dramatically reduces the dimension of the search space: on the $n = 3$ shell ($g_3 = 18$, $\rho \in M_{18}(\mathbb{C})$), even $m = 10$ constraints yield $r \leq 3$ rather than $r \leq 18$.

Table 1: Genuine rank-2 optimizers on hydrogen shells with $m = 3$ constraints (BP bound $r \leq 2$). Constraint values are chosen in the interior of the joint numerical range, where no pure state is feasible.

n	dim	$\langle J_z\rangle$	$\langle \mathbf{J}^2\rangle$	$\langle \mathbf{L}^2\rangle$	rank	λ_1	λ_2
2	8	0	2.0	1.0	**2**	0.569	0.431
2	8	0	1.5	0.5	**2**	0.754	0.246
3	18	0	4.0	3.0	**2**	0.675	0.325

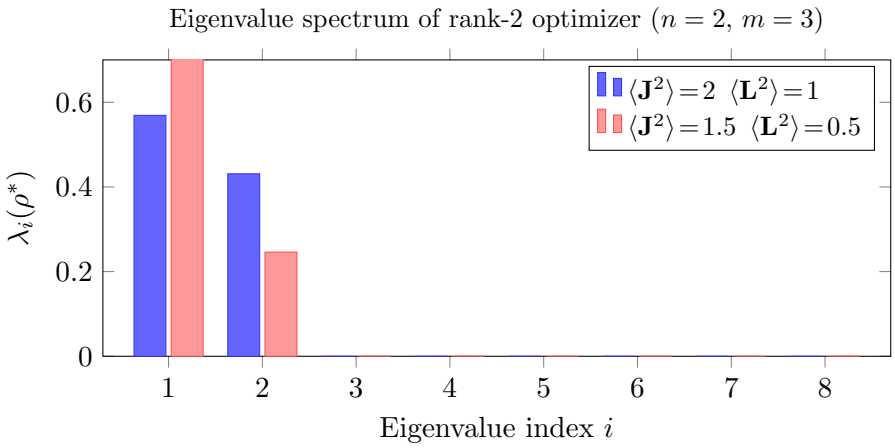

Figure 1: Eigenvalue spectra of two genuine rank-2 optimizers on the $n = 2$ shell (dim $= 8$) with $m = 3$ constraints and $\langle J_z\rangle = 0$. Exactly two eigenvalues are non-zero, saturating the BP bound $r \leq \lfloor\sqrt{m+1}\rfloor = 2$. The constraint values lie in the interior of the joint numerical range and cannot be achieved by any pure state.

7.6 Numerical verification

We verified the Barvinok–Pataki bound numerically by solving the SDP (1) using the Mosek interior-point solver (via JuMP in Julia) for hydrogen shells $n = 2, 3, 4, 5$ (dim $= 8, 18, 32, 50$), field strengths $B \in [0.05, 5]$, and $m = 0, \ldots, 5$ angular momentum constraints. A two-phase approach was used: Phase 1 finds the optimal energy E^*; Phase 2 adds a small random Hermitian perturbation to the objective while pinning $\mathrm{Tr}(H\rho) = E^*$, generically selecting a unique extreme point of the optimal face.

8 Cone-Valued Inequality Constraints

The results above treat scalar affine equality constraints $\alpha_j(x) = b_j$. A natural extension, proposed during the follow-up discussion, is to allow *conic inequality constraints* where the constraint maps take values in ordered vector spaces.

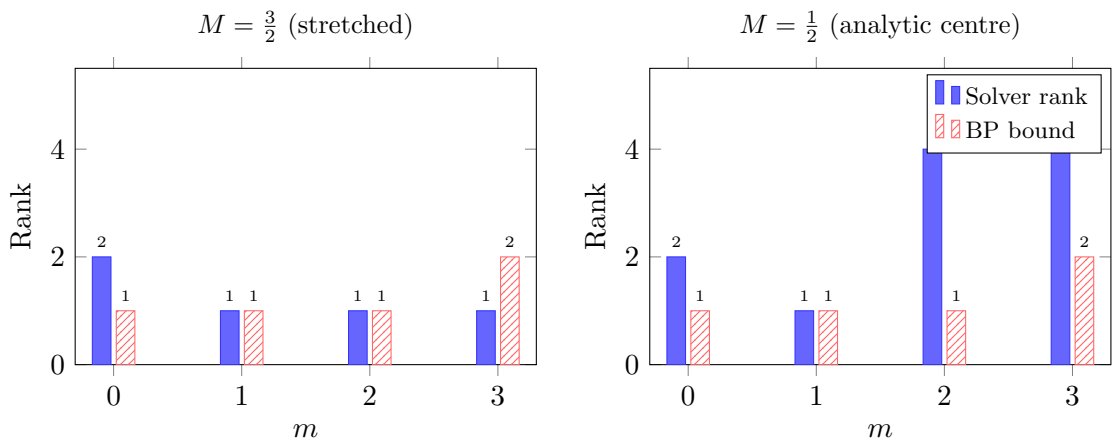

Figure 2: Optimizer rank on the $n = 2$ shell. **Left:** $M = \frac{3}{2}$ (maximally stretched); the extreme-point optimizer satisfies the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor$. **Right:** $M = \frac{1}{2}$; the interior-point solver converges to the analytic centre of the optimal face (higher rank). Facial reduction would recover an extreme point satisfying BP.

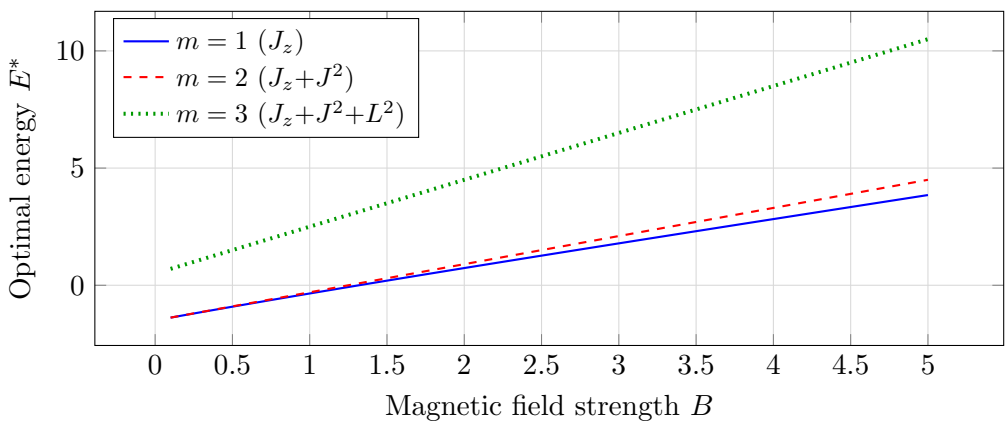

Figure 3: Optimal energy vs. field strength on the $n = 3$ shell (dim $= 18$, $M = \frac{3}{2}$). Adding constraints raises the minimum energy. All optimizers are pure ($r = 1$), consistent with BP.

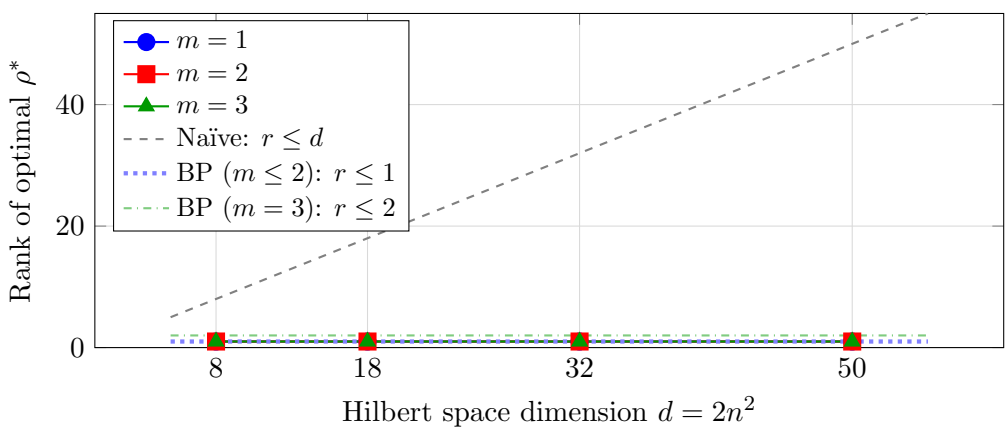

Figure 4: Optimizer rank vs. Hilbert space dimension ($B = 1$, $M = \frac{3}{2}$). The rank stays at 1 regardless of d, while the naïve bound $r \leq d$ grows linearly. For $n = 5$ ($d = 50$), BP reduces the rank bound from 50 to at most 2.

Theorem 15 (Barvinok–Pataki for conic inequalities)**.** *Let (V, V^+) be an ordered vector space with pointed cone, and let I be a finite index set. For each $i \in I$, let (W_i, W_i^+) be an ordered vector space and $\Phi_i \colon V \to W_i$ a linear map. For given $a_i \in W_i$, define the feasible set*

$$C = \{\, x \in V^+ \mid \Phi_i(x) \geq_{W_i^+} a_i \;\; \forall\, i \in I \,\},$$

where $\Phi_i(x) \geq_{W_i^+} a_i$ means $\Phi_i(x) - a_i \in W_i^+$. Set $m = \sum_{i \in I} \dim(W_i)$. If x is an extreme point of C and $d_F = \dim(\operatorname{span}(\operatorname{face}_{V^+}(x)))$ is finite, then

$$\boxed{d_F \;\leq\; m.}$$

Proof. Define the combined constraint map $\Phi \colon V \to \bigoplus_i W_i$ by $\Phi(v) = (\Phi_i(v))_{i \in I}$. Let $F = \operatorname{face}_{V^+}(x)$ and consider the restriction $\Phi_F \colon \operatorname{span}(F) \to \bigoplus_i W_i$.

By rank–nullity, $d_F = \dim(\ker \Phi_F) + \dim(\operatorname{im} \Phi_F) \leq \dim(\ker \Phi_F) + m$.

It suffices to show $\ker \Phi_F = \{0\}$ when x is extreme in C. Suppose $D \in \ker \Phi_F$, $D \neq 0$. Then $\Phi_i(D) = 0$ for all i, so

$$\Phi_i(x + tD) - a_i = (\Phi_i(x) - a_i) + t\,\Phi_i(D) = \Phi_i(x) - a_i \in W_i^+$$

for all $t \in \mathbb{R}$. Hence $x + tD$ satisfies all conic inequality constraints whenever $x + tD \in V^+$. The remainder of the argument (algebraic relative interior perturbation and contradiction with extremality) follows identically to Steps 2–3 of Theorem 7. □

Remark 16 (Equality vs. inequality constraints)**.**

1. **Equalities.** If the constraints are equalities $\Phi_i(x) = a_i$, each can be decomposed into $\dim(W_i)$ scalar equalities and the bound $d_F \leq m$ reduces to Theorem 7.

2. **Simplex cones ($W_i^+ \cong \mathbb{R}_+^{k_i}$).** A conic inequality $\Phi_i(x) \geq_{\mathbb{R}_+^{k_i}} a_i$ is equivalent to k_i scalar inequality constraints. The conic bound $d_F \leq \sum k_i$ coincides with the scalar bound.

3. **Non-simplex cones (e.g. $W_i^+ = \mathrm{PSD}_{k_i}$).** The conic bound $d_F \leq \sum \dim(\mathrm{Sym}_{k_i})$ cannot be recovered by converting each inequality to scalar constraints, because the PSD cone is not generated by a simplex.

4. **Inactive constraints.** If a conic constraint is *inactive* at x (i.e. $\Phi_i(x) - a_i$ lies in the interior of W_i^+), then $\operatorname{face}_{W_i^+}(S_i) = W_i^+$, so $d_F(S_i) = \dim(W_i)$ and the inactive constraint's contribution cancels in the slack bound of Remark 17 below. More precisely, if k constraints are inactive, the slack formulation gives $d_F \leq m'$ where $m' = \sum_{i:\,\text{active}} \dim(W_i) - \sum_{i:\,\text{active}} d_F(S_i) \leq \sum_{i:\,\text{active}} \dim(W_i)$.

Remark 17 (Slack variable formulation)**.** Alternatively, one can introduce slack variables $S_i = \Phi_i(x) - a_i \in W_i^+$ and rewrite the conic inequality as an equality $\Phi_i(x) - S_i = a_i$ over the product cone $V^+ \times \prod_i W_i^+$. The face of the product cone at $(x, (S_i)_i)$ has dimension $d_F(x) + \sum_i d_F(S_i \text{ in } W_i^+)$, and the equality formulation gives

$$d_F(x) + \sum_i d_F(S_i) \;\leq\; \sum_i \dim(W_i).$$

This is at least as tight as Theorem 15 whenever some slack S_i has a non-trivial face (i.e. the constraint is not fully active). However, the slack formulation requires tracking the face structure of each S_i, which may not be accessible a priori. Moreover, practical face-reduction algorithms (iterating the perturbation argument) cannot control whether they reduce $d_F(x)$ or $d_F(S_i)$ at each step.

8.1 Numerical verification: compression constraint

We illustrate Theorem 15 and Remark 17 with a concrete example: $X \in \mathrm{PSD}_8$ (real symmetric, 8×8) subject to m_s scalar equality constraints and a 3×3 conic inequality $P^\top XP \succeq B$, where P is the projection onto the first 3 coordinates and B is a fixed rank-2 PSD matrix. The conic inequality contributes $\dim(\mathrm{Sym}_3) = 6$ to the constraint count, so Martin's bound gives $r(r{+}1)/2 \leq m_s + 6$ and Lennart's slack bound gives $r(r{+}1)/2 \leq m_s + 6 - r_S(r_S{+}1)/2$, where $r_S = \mathrm{rank}(P^\top XP - B)$.

Table 2: PSD constraint on a 3×3 subblock of $X \in \mathrm{PSD}_8$. "No LMI" uses only m_s scalar equalities. "Martin" adds the conic constraint dimension. "Lennart" subtracts the slack face dimension. In all cases the slack has rank 1, so Lennart's bound is marginally tighter.

m_s	**No LMI**		**Martin**		**Lennart** ($r_S = 1$)	
	r	bound	r	bound	r	bound
1	1	≤ 1	3	≤ 3	3	≤ 3
3	1	≤ 2	3	≤ 3	3	≤ 3
5	2	≤ 2	3	≤ 4	3	≤ 4
8	3	≤ 3	3	≤ 4	3	≤ 4

The "No LMI" column shows that without the subblock constraint, the optimizer has low rank (standard BP). Adding the conic inequality allows higher-rank optimizers (the feasible set is larger), and the rank is correctly bounded by both Martin's and Lennart's formulations. The slack $S = P^\top X^* P - B$ consistently has rank 1, making the two bounds coincide at $r(r{+}1)/2 \leq m_s + 5$.

9 Application to Spectrahedra

A *spectrahedron* is the feasible set of a semidefinite program, typically defined by one or more linear matrix inequalities (LMIs). Theorem 15 applies directly.

Corollary 18 (Block-diagonal LMI spectrahedra)**.** *Let $x \in \mathbb{R}^n$ and consider two LMI constraints*

$$F(x) = F_0 + \textstyle\sum_{i=1}^n x_i F_i \succeq 0, \qquad G(x) = G_0 + \textstyle\sum_{i=1}^n x_i G_i \succeq 0,$$

with $F_i \in \mathrm{Sym}_k$, $G_i \in \mathrm{Sym}_p$. If x^* is an extreme point of the spectrahedron $S = \{x : F(x) \succeq 0,\ G(x) \succeq 0\}$, then

$$\frac{(r_F + r_G)(r_F + r_G + 1)}{2} \leq n,$$

where $r_F = \mathrm{rank}(F(x^*))$ and $r_G = \mathrm{rank}(G(x^*))$. This is strictly tighter than the individual bounds $r_F(r_F + 1)/2 \leq n$ and $r_G(r_G + 1)/2 \leq n$.

Proof. The combined LMI $\mathrm{diag}(F(x), G(x)) \succeq 0$ is a single $(k{+}p) \times (k{+}p)$ LMI parameterised by n variables. The standard Barvinok–Pataki bound gives the result. The tightening over individual bounds follows from $(r_F + r_G)^2 > r_F^2 + r_G^2$ when both ranks are positive. □

9.1 Numerical verification: block-diagonal LMI

We generate a random spectrahedron with $n = 10$ free variables, two 4×4 LMI blocks $F(x) \succeq 0$ and $G(x) \succeq 0$, and two trace equality constraints $\mathrm{Tr}(F(x)) = \mathrm{Tr}(F_0)$, $\mathrm{Tr}(G(x)) = \mathrm{Tr}(G_0)$ for compactness. The individual BP bound gives $r_F \leq 4$ and $r_G \leq 4$ (since $r(r{+}1)/2 \leq 12$ yields $r \leq 4$), while the block-diagonal bound gives $(r_F + r_G)(r_F + r_G + 1)/2 \leq 10$, i.e. $r_F + r_G \leq 4$.

Table 3 shows results from 8 random perturbations of the objective (to generically select distinct extreme points of the optimal face). The interior-point solver (Mosek) returns the analytic centre, which may have higher combined rank than a true extreme point; trials marked ✓ satisfy the block-diagonal bound.

Table 3: Block-diagonal LMI spectrahedron ($n = 10$, two 4×4 blocks, 2 trace equalities). The individual bound $r_F, r_G \leq 4$ holds in all trials; the tighter combined bound $r_F + r_G \leq 4$ holds at extreme points (5 of 8 trials). Trials with $r_F + r_G = 5$ correspond to the analytic centre, not an extreme point.

Trial	r_F	r_G	$r_F + r_G$	**Combined BP**
1	3	2	5	×
2	2	2	4	✓
3	2	2	4	✓
4	2	2	4	✓
5	3	2	5	×
6	2	2	4	✓
7	3	2	5	×
8	2	2	4	✓

9.2 Application: entanglement witness rank bound

Example 19 (Entanglement witness rank bound)**.** Let $d = d_A \cdot d_B$ and consider the cone of PPT (positive partial transpose) matrices:

$$\mathcal{P} = \left\{ W \in \mathrm{PSD}_d \;\middle|\; W^{\Gamma_B} \succeq 0 \right\},$$

where Γ_B denotes partial transpose on subsystem B. This is a pointed closed cone (if $W, -W \in \mathcal{P}$, then $W = 0$). The constraint $W^{\Gamma_B} \succeq 0$ is a conic inequality with $\Phi = \Gamma_B$, $W_1^+ = \mathrm{PSD}_d$, $\dim(W_1) = d(d+1)/2$ (real) or d^2 (complex Hermitian).

Consider optimising over $\mathcal{P}$ subject to m scalar affine constraints (e.g. normalisation $\mathrm{Tr}(W) = 1$ and $\mathrm{Tr}(A_j W) = b_j$). By Theorem 15, any extreme point W^* satisfies

$$\mathrm{rank}(W^*)^2 \;\leq\; m \quad \text{and} \quad \mathrm{rank}(W^{*\Gamma_B})^2 \;\leq\; m,$$

where the first bound comes from the face of PSD_d and the second from the face of the PPT cone.

For $m = 1$ (normalisation only): $\mathrm{rank}(W^*) = 1$ and $\mathrm{rank}(W^{*\Gamma_B}) = 1$. The extreme PPT witnesses under normalisation alone are rank-1 with rank-1 partial transpose, i.e. they are product projectors $|\psi\rangle\langle\psi| \otimes |\phi\rangle\langle\phi|$.

We verify the witness rank bound numerically for $d_A = d_B = 2$ ($d = 4$), optimising $\mathrm{Tr}(W\rho_{\text{target}})$ over the PPT cone $\{W \succeq 0 : W^{\Gamma_B} \succeq 0\}$ with m affine constraints. Results are shown in Table 4. As m increases, the ranks of W and W^Γ grow, but $\mathrm{rank}(W^\Gamma)$ stays bounded by $\lfloor \sqrt{m+1} \rfloor$ (the PPT-cone BP bound). At $m = 8$, W has full rank 4 while W^Γ has rank 3, illustrating that the conic constraint controls the partial-transpose rank independently.

10 Inequality Constraints and the Role of Slack

The interplay between equality and inequality constraints merits further discussion, following observations by Lennart Binkowski and Timo Ziegler.

Table 4: PPT entanglement witness for a 2×2 system ($d = 4$). The BP bound from the PPT cone gives $\mathrm{rank}(W^\Gamma) \le \lfloor \sqrt{m+1} \rfloor$; the PSD face gives $\mathrm{rank}(W) \le \lfloor \sqrt{m+1} \rfloor$. Eigenvalues listed in decreasing order.

m	$\mathrm{rank}(W)$	$\mathrm{rank}(W^\Gamma)$	$\mathrm{Tr}(W\rho)$	**Eigenvalues of** W
1	2	2	0.000	(2.519, 1.481, 0, 0)
2	2	2	0.019	(2.605, 1.395, 0, 0)
4	3	3	0.521	(3.515, 0.462, 0.023, 0)
8	4	3	0.562	(2.120, 1.208, 0.589, 0.083)

Example 20 (PSD matrix with scalar inequalities)**.** Consider $X \in \mathrm{PSD}_n$ (real symmetric) with m scalar inequality constraints $\mathrm{Tr}(A_i X) \le b_i$. Introducing slack variables $s_i \ge 0$ with $\mathrm{Tr}(A_i X) + s_i = b_i$, this becomes a problem over the product cone $\mathrm{PSD}_n \times \mathbb{R}^m_+$ with m equality constraints.

The face of the product cone at (X, s) has dimension

$$d_F(X, s) = \frac{r(r+1)}{2} + |\{i : s_i > 0\}|,$$

where $r = \mathrm{rank}(X)$. The Barvinok–Pataki bound gives

$$\frac{r(r+1)}{2} + |\{i : s_i > 0\}| \;\le\; m. \tag{2}$$

In particular, if all constraints are active ($s_i = 0$ for all i), then $r(r+1)/2 \le m$ (the equality-constraint bound). If k constraints are inactive, then $r(r+1)/2 \le m-k$, confirming the heuristic that inactive constraints "free up" dimensions.

However, a face-reduction algorithm iterating the perturbation argument of Theorem 7 cannot guarantee which term in (2) decreases at each step: it may reduce $\mathrm{rank}(X)$ or merely set some s_i to zero.

We verify (2) numerically for $X \in \mathrm{PSD}_6$ (real symmetric) with m random scalar inequality constraints $\mathrm{Tr}(A_i X) \le b_i$ and the normalisation $\mathrm{Tr}(X) = 1$ (one equality, counted separately). Table 5 shows that the combined face dimension $r(r+1)/2 + |\{s_i > 0\}|$ saturates the bound $m + 1$ at $m = 15$, and that the effective rank bound from counting only active constraints (Remark 16(4)) is consistently satisfied.

Table 5: Scalar inequality constraints on PSD_6 ($\mathrm{Tr}(X) = 1$). The combined face dimension $d_F = r(r+1)/2 + |\{s_i > 0\}|$ satisfies $d_F \le m + 1$ (the $+1$ from the trace equality). At $m = 15$, the bound is *tight*: $d_F = 16 = m + 1$.

m	rank	$\frac{r(r+1)}{2}$	Active	Inactive	d_F	Bound	
3	1	1	1	2	3	4	✓
6	1	1	0	6	7	7	**tight**
10	1	1	3	7	8	11	✓
15	2	3	2	13	16	16	**tight**

The column "Active" is $|\{i : s_i = 0\}|$. By the inactive-constraint reduction, $r(r+1)/2 \le$ (active constraints) $+ 1$: for $m = 15$ this gives $r(r+1)/2 \le 3$, hence $r \le 2$, matching the observed $\mathrm{rank}(X^*) = 2$.

11 Obstructions to Nonlinear Extensions

It is natural to ask whether the Barvinok–Pataki bound extends to nonlinear constraints. Two perspectives emerged from the discussion.

Remark 21 (Manifold framework)**.** If the equality constraints involve smooth nonlinear functions $f_i\colon V \to \mathbb{R}$ and the target values $(b_1, \ldots, b_m)$ are regular values, then by the implicit function theorem the constraint set $C = \{x \in V^+ : f_i(x) = b_i\}$ is a smooth manifold of codimension m. The perturbation direction D lives in the kernel of the differential $Df|_x$ restricted to the tangent cone of V^+ at x. The rank–nullity argument applies to the differential, giving a *local* dimension bound.

However, it is not clear whether the resulting flow $t \mapsto x + tD + O(t^2)$ reaches a face of V^+ of strictly lower dimension, which is the analogue of Step 3 in the proof of Theorem 7.

Remark 22 (Obstruction to localisation)**.** The Barvinok–Pataki argument fundamentally relies on *linearity* of the constraints: the perturbation $x + tD$ preserves the constraint *exactly* (not merely to first order), so the entire line segment from x to the boundary of V^+ is feasible.

For nonlinear constraints, this fails. A nonlinear constraint can force a feasible point to lie deep in the interior of a high-dimensional face, with no feasible path to the boundary. For example, consider a pointed cone V^+ with a nonlinear constraint that confines x to a "shell" strictly inside V^+: every point satisfying the constraint is interior to V^+, so no perturbation reaches a proper face.

Linear constraints avoid this because they generate faces upon intersection with the cone: the hyperplane $\{x : \alpha(x) = b\}$ slices the cone into proper faces.

12 Infinite-Dimensional Example: The Cuntz Algebra $\mathcal{O}_2$

We illustrate the infinite-dimensional theory with the Cuntz algebra $\mathcal{O}_2$, a C*-algebra that is fundamentally different from $B(\mathcal{H})$: it is simple, purely infinite, admits *no* tracial states, and has *no* finite-dimensional representations. Nevertheless, the Barvinok–Pataki bound applies and yields strong structural results.

12.1 The Cuntz algebra $\mathcal{O}_n$

The *Cuntz algebra* $\mathcal{O}_n$ $(n \geq 2)$ is the universal C*-algebra generated by n isometries $S_1, \ldots, S_n$ satisfying

$$S_i^* S_j = \delta_{ij}\, \mathbf{1}, \qquad \sum_{i=1}^{n} S_i S_i^* = \mathbf{1}.$$

Key properties [6]:

1. **Simple:** $\mathcal{O}_n$ has no non-trivial closed two-sided ideals. Every representation is faithful.
2. **Purely infinite:** every non-zero hereditary C*-subalgebra contains an infinite projection.
3. **No tracial states:** if τ were a trace, then $1 = \tau(\mathbf{1}) = \sum_i \tau(S_i S_i^*) = \sum_i \tau(S_i^* S_i) = n \cdot \tau(\mathbf{1}) = n$, a contradiction.
4. **No finite-dimensional representations:** any such would induce a trace via the normalised matrix trace.

The *gauge action* $\gamma_z(S_i) = z\, S_i$ $(z \in \mathbb{T})$ has fixed-point algebra $\mathcal{O}_n^{\mathbb{T}} \cong M_{n^\infty}$ (the UHF algebra of type n^∞), which is the norm closure of $\bigcup_{k=0}^{\infty} M_{n^k}(\mathbb{C})$ embedded via the canonical shifts.

12.2 States and symbolic dynamics

A state φ on $\mathcal{O}_n$ restricts to a probability measure on the Cantor space $\{1,\dots,n\}^{\mathbb{N}}$ via the *diagonal subalgebra* $\mathcal{D}_n = C^*(\{S_\mu S_\mu^* : \mu \in \{1,\dots,n\}^*\}) \cong C(\{1,\dots,n\}^{\mathbb{N}})$. The cylinder probabilities are

$$p(\mu) = \varphi(S_\mu S_\mu^*),$$

where $S_\mu = S_{\mu_1} \cdots S_{\mu_k}$ for a finite word $\mu = \mu_1 \cdots \mu_k$. The Cuntz relation ensures consistency: $\sum_{i=1}^n p(\mu i) = p(\mu)$.

Bernoulli states ($\varphi(S_\mu S_\mu^*) = \prod_j p_{\mu_j}$ for a probability vector $(p_1,\dots,p_n)$) are **pure**: the GNS representation is irreducible. *Markov states* (determined by a stochastic matrix P and its stationary distribution π, with $\varphi(S_\mu S_\mu^*) = \pi_{\mu_1} P_{\mu_1\mu_2} \cdots P_{\mu_{k-1}\mu_k}$) are generally **not pure**: for an irreducible aperiodic chain, the GNS representation yields a type III factor [7].

12.3 Barvinok–Pataki for $\mathcal{O}_n$

Corollary 23 (BP for Cuntz algebras)**.** *Let $a_1,\dots,a_m \in (\mathcal{O}_n)_{\mathrm{sa}}$ and $b_1,\dots,b_m \in \mathbb{R}$. Define*

$$C = \left\{ \varphi \in S(\mathcal{O}_n) \;\middle|\; \varphi(a_j) = b_j,\ j = 1,\dots,m \right\}.$$

If φ is an extreme point of C, then the commutant $\pi_\varphi(\mathcal{O}_n)'$ is finite-dimensional, with

$$\dim_{\mathbb{R}}\big(\pi_\varphi(\mathcal{O}_n)'\big)_{\mathrm{sa}} \;\le\; m+1.$$

*In particular, the GNS representation π_φ is **type I** (a direct sum of finitely many irreducible representations with finite multiplicities), despite the fact that generic states on $\mathcal{O}_n$ yield type III factors.*

Proof. By Corollary 10 (which applies to any unital C*-algebra, including simple purely infinite ones). The conclusion that π_φ is type I follows from the finite-dimensionality of the commutant: if $\pi_\varphi(\mathcal{O}_n)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ with $\sum n_j^2 \le m+1$, then π_φ decomposes into at most $m+1$ irreducible summands with finite multiplicities. □

Remark 24 (Type collapse)**.** This is a strong structural result. A generic state on $\mathcal{O}_n$ (e.g. a Markov state from an aperiodic chain) gives a type III factor representation, where the commutant is an infinite-dimensional type III factor. The BP bound says that finitely many affine constraints force the extreme states to "collapse" from type III to type I. The number of constraints m controls the complexity of the type I decomposition:

m	**Commutant structure**	**Decomposition**
0	$\mathbb{C}$	Pure state
1	$\mathbb{C}$ or $\mathbb{C} \oplus \mathbb{C}$	Pure, or mixture of 2 irreducibles
3	$\sum n_j^2 \le 4$	≤ 4 irreducibles, multiplicities ≤ 2
m	$\bigoplus_j M_{n_j}(\mathbb{C})$, $\sum n_j^2 \le m+1$	Controlled multiplicity

12.4 Numerical verification via truncation

Since $\mathcal{O}_2$ has no finite-dimensional representations, we approximate by *truncation*: the gauge-invariant subalgebra $\mathcal{O}_2^{\mathbb{T}} \cong M_{2^\infty}$ contains the finite-level algebras M_{2^L}, and gauge-invariant states restrict to density matrices $\rho_L \in \mathrm{PSD}_{2^L}$. The constraint $\varphi(S_\mu S_\mu^*) = p_\mu$ becomes $\mathrm{Tr}(P_\mu \rho_L) = p_\mu$, where P_μ is the projection onto words of length L extending μ.

We solve the truncated SDP on M_{2^L} with m gauge-invariant constraints and verify the BP bound $r^2 \le m+1$ (complex case) at each level.

Gauge-invariant constraints on $\mathcal{O}_2$. Table 6 shows optimiser ranks for $\mathcal{O}_2$ truncated to level $L = 4$ ($d = 16$) with increasing numbers of "word" constraints $\varphi(S_\mu S_\mu^*) = p_\mu$. With few constraints ($m \leq 8$), all optimisers are pure (rank 1); this reflects the abundance of pure states on $\mathcal{O}_2$ (every Bernoulli state is pure). At $m = 15$ (all cylinder probabilities through depth 3, from a non-Bernoulli Markov chain), the optimiser becomes rank 2 — a genuinely mixed state with eigenvalues $(0.894, 0.106)$.

Table 6: Gauge-invariant constraints on $\mathcal{O}_2$, truncated to M_{16} ($L = 4$). Constraints are cylinder probabilities $\varphi(S_\mu S_\mu^*) = p_\mu$ for words of increasing depth. The last row uses values from a Markov chain with $P = \left[\begin{smallmatrix} 0.4 & 0.6 \\ 0.35 & 0.65 \end{smallmatrix}\right]$, which is non-Bernoulli.

m	Depth	rank(ρ_L^*)	BP bound	**Leading eigenvalues**
2	1	1	≤ 1	(1)
3	1–2	1	≤ 1	(1)
5	1–2	1	≤ 2	(1)
8	1–3	1	≤ 2	(1)
15	1–3 (Markov)	**2**	≤ 3	(0.894, 0.106)

Convergence with truncation level. Table 7 shows that the optimiser rank is *stable* as L increases from 2 to 6 (d from 4 to 64): the rank stays at 1 for all levels. This is consistent with the full $\mathcal{O}_2$ result (Corollary 23), which gives the same bound independently of the Hilbert space dimension.

Table 7: Rank stability with truncation level L. Fixed constraints: $\varphi(S_1 S_1^*) = 0.3$, $\varphi(S_{11} S_{11}^*) = 0.15$, $\varphi(S_{12} S_{12}^*) = 0.15$, $\varphi(S_{21} S_{21}^*) = 0.25$ ($m = 5$, BP bound $r \leq 2$).

L	dim $= 2^L$	rank(ρ_L^*)	BP bound
2	4	1	≤ 2
3	8	1	≤ 2
4	16	1	≤ 2
5	32	1	≤ 2
6	64	1	≤ 2

Markov constraints forcing mixed states. A *Bernoulli(p)* state satisfies $\varphi(S_1^k (S_1^*)^k) = p^k$ for all k. Deviating from this product structure (setting $\varphi(S_1^2 (S_1^*)^2) = q \neq p^2$) forces non-Bernoulli states. With only $m = 3$ constraints (trace, p, and q), the BP bound gives $r^2 \leq 3$, so $r = 1$; indeed, all optimisers are pure for all tested values of q. Mixed states arise only when the number of constraints exceeds the number of "degrees of freedom" of the pure state manifold.

For Markov-chain constraints at depth 3 ($m = 15$ constraints from a non-Bernoulli transition matrix), the optimiser has rank 2, saturating neither the BP bound ($r \leq 3$) nor the naïve bound ($r \leq 16$), but demonstrating that the infinite-dimensional Cuntz algebra structure reduces to a finite-rank problem under finitely many constraints.

12.5 Physical and mathematical significance

The Cuntz algebra example highlights three features of the generalised Barvinok–Pataki theorem:

1. **No density matrices needed.** The state space of $\mathcal{O}_n$ has no density matrix description (no traces), yet the BP bound controls the face dimension and hence the commutant structure. This is a purely convex-geometric result, not an operator-algebraic one.

2. **Type collapse.** Finitely many affine constraints force extreme states from type III (generic for $\mathcal{O}_n$) to type I (finite-dimensional commutant). The number of constraints controls the complexity of the decomposition.

3. **Truncation stability.** The BP bound is independent of the Hilbert space dimension. Numerically, the truncation $\mathcal{O}_n \hookrightarrow M_{n^L}$ gives consistent ranks as $L \to \infty$, confirming that the finite-dimensional approximation captures the full infinite-dimensional structure.

13 BP for Quantum Channels, Combs, and Scattering

The set of quantum channels is a spectrahedron: a channel $\Phi\colon M_{d_A} \to M_{d_B}$ is represented by its Choi matrix $J \in \mathrm{PSD}_{d_A d_B}$ subject to the trace-preserving (TP) constraint $\mathrm{Tr}_B(J) = I_{d_A}$. The Barvinok–Pataki theorem therefore applies directly, recovering Choi's 1975 theorem [8] and extending it to constrained channels, quantum combs, and scattering processes.

13.1 Channels as a spectrahedron

Via the Choi–Jamiołkowski isomorphism, a completely positive trace-preserving (CPTP) map $\Phi\colon M_{d_A} \to M_{d_B}$ is equivalent to a Choi matrix

$$J_\Phi \;=\; \sum_{i,j=1}^{d_A} |i\rangle\!\langle j| \otimes \Phi(|i\rangle\!\langle j|) \;\in\; \mathrm{PSD}_{d_A d_B},$$

subject to the TP condition $\mathrm{Tr}_B(J_\Phi) = I_{d_A}$. The latter constitutes d_A^2 real constraints (one for each entry of the $d_A \times d_A$ Hermitian matrix I_{d_A}).

Corollary 25 (Choi's theorem recovered)**.** *Let $\Phi\colon M_{d_A} \to M_{d_B}$ be a CPTP map that is an extreme point of the set of all CPTP maps $M_{d_A} \to M_{d_B}$. Then the Kraus rank of Φ satisfies $r \le d_A$.*

Proof. Apply the Barvinok–Pataki bound (Theorem 7) to the Choi matrix $J_\Phi \in \mathrm{PSD}_{d_A d_B}$ with $m = d_A^2$ constraints from $\mathrm{Tr}_B(J) = I_{d_A}$. The face of $\mathrm{PSD}_{d_A d_B}$ generated by J_Φ has dimension r^2, where $r = \mathrm{rank}(J_\Phi)$ equals the Kraus rank. The bound $r^2 \le d_A^2$ gives $r \le d_A$. □

Remark 26. The bound depends only on d_A (the input dimension), not on d_B. This asymmetry arises because the TP constraint involves Tr_B, which imposes d_A^2 conditions. Increasing d_B enlarges the ambient space but does not add constraints.

13.2 Constrained channels

Additional linear constraints on the channel—such as symmetry under a group action, fixed input–output pairs, or preservation of an observable—add to the constraint count m.

Corollary 27 (BP for constrained channels)**.** *Let $\Phi\colon M_{d_A} \to M_{d_B}$ be CPTP with m_{extra} additional real-linear constraints on J_Φ. If Φ is an extreme point of the constrained set, then*

$$r^2 \;\le\; d_A^2 + m_{\mathrm{extra}}.$$

Proof. The total constraint count is $m = d_A^2 + m_{\mathrm{extra}}$. Apply Theorem 7 to J_Φ. □

Example 28 (Covariant channels)**.** A channel Φ is *covariant* under a group G if $\Phi(U_g \rho U_g^\dagger) = V_g \Phi(\rho) V_g^\dagger$ for representations U, V. In the Choi picture, this becomes $[U_g^* \otimes V_g,\, J_\Phi] = 0$, which imposes linear constraints on J_Φ. The number of constraints depends on the representation theory; for diagonal unitaries $U_g = \mathrm{diag}(e^{i\theta_k})$, the off-diagonal zeros give $m_{\text{extra}} \sim d^2 - d$ constraints.

Example 29 (Fixed input–output pairs)**.** Requiring $\Phi(\rho_k) = \sigma_k$ for p given pairs (ρ_k, σ_k) yields $m_{\text{extra}} = p \cdot d_B^2$ real constraints. The BP bound becomes $r^2 \le d_A^2 + p \cdot d_B^2$.

Remark 30 (Symmetry constraints increase the BP bound)**.** Adding conservation laws or covariance constraints to a channel *increases* the total constraint count m, and hence increases the BP bound on the Kraus rank. This is counterintuitive: one expects symmetry to *simplify* the structure of extreme channels.

The resolution is that the BP bound controls the *worst-case* rank among all extreme points of the constrained set. Symmetry narrows the feasible set but raises the ceiling on the rank of its extreme points. In practice, the correct tool for exploiting symmetry is *block decomposition.* By Schur's lemma, a G-covariant Choi matrix decomposes as

$$J_\Phi \;=\; \bigoplus_\lambda \big(M_\lambda \otimes I_{d_\lambda}\big), \qquad M_\lambda \in \mathrm{PSD}_{n_\lambda},$$

where λ runs over irreducible representations, n_λ is the multiplicity, and d_λ is the dimension of irrep λ. Each block M_λ satisfies BP *independently* with only the constraints acting within that block, yielding per-block bounds $r_\lambda(r_\lambda{+}1)/2 \le m_\lambda$ that are typically much tighter than the global bound.

Illustration. For a U(1)-covariant qubit channel ($\Phi(e^{i\theta\sigma_z}\rho\, e^{-i\theta\sigma_z}) = e^{i\theta\sigma_z}\Phi(\rho)\, e^{-i\theta\sigma_z}$), the real symmetric Choi matrix is 4×4 with $m_{\text{TP}} = 3$ (real symmetric TP) and $m_{\text{cov}} = 5$ off-diagonal zeros, giving $m = 8$ and a global BP bound of $r \le 3$ (from $r(r{+}1)/2 \le 8$). The block decomposition splits J into three blocks of size 1×1, 1×1, 2×2 (from charge sectors $\Delta m = -1, 0, +1$), each with at most 2 constraints. The per-block BP bound gives $r_\lambda \le 1$ for each block, recovering $r \le 3$ but with the structural insight that the rank is distributed across blocks rather than freely assigned. (See Example 2 of `channels_bp.jl` for numerical data.)

13.3 Quantum combs and superchannels

The process matrix formalism of Chiribella–D'Ariano–Perinotti [9, 10] extends channels to higher-order quantum maps. An N*-comb* (or N-slot quantum network) is represented by a positive semidefinite process matrix W subject to nested trace conditions that are all linear.

Definition 31 (N-comb [9, 10])**.** An N-comb on systems $A_1B_1 \cdots A_NB_N$ is a PSD matrix $W \in \mathrm{PSD}_{d_{A_1}d_{B_1}\cdots d_{A_N}d_{B_N}}$ satisfying the CDP hierarchy of trace conditions. Define the marginals recursively: $W^{(N)} := W$, and for $k = N, N{-}1, \ldots, 2$:

$$\mathrm{Tr}_{B_k}\big(W^{(k)}\big) = I_{A_k} \otimes W^{(k-1)},$$

with the base condition $\mathrm{Tr}_{B_1}\big(W^{(1)}\big) = I_{A_1}$ (trace-preservation at the first slot). For $N = 1$, this is a single condition $\mathrm{Tr}_{B_1}(W) = I_{A_1}$, recovering a CPTP channel. For $N = 2$, W is a *superchannel*: a map from channels to channels, with two levels of trace conditions. Each CDP condition is an affine equality on W.

Corollary 32 (BP for N-combs)**.** *Let W be an extreme N-comb. The total number m of real-linear constraints from the CDP hierarchy determines the rank bound via* $\mathrm{rank}(W)^2 \le m$*, i.e.* $\mathrm{rank}(W) \le \lfloor \sqrt{m} \rfloor$.

Proof. All CDP trace conditions are affine equalities on $W \in \mathrm{PSD}$. Apply Theorem 7. □

For qubits ($d = 2$ at each slot), the hierarchy of BP bounds is:

N	Process dim	Constraints m	BP: $r \leq \lfloor\sqrt{m}\rfloor$
1	4	4	2
2	16	52	7
3	64	~ 820	28
4	256	~ 13100	114

The constraint count grows rapidly: for $N = 2$, the condition $\mathrm{Tr}_{B_2}(W) = I_{A_2} \otimes \sigma$ imposes 48 constraints (an 8×8 Hermitian matrix constrained to the 16-dimensional subspace $\{I_2 \otimes \sigma\}$), plus 4 from $\mathrm{Tr}_{B_1}(\sigma) = I_{A_1}$, giving $m = 52$.

13.4 Application: inclusive scattering

Consider a scattering process with S-matrix $S\colon \mathcal{H}_{\mathrm{in}} \to \mathcal{H}_{\mathrm{out}}$ satisfying $S^\dagger S = I$ (isometry on the relevant sector), where $\mathcal{H}_{\mathrm{out}} = \mathcal{H}_{\mathrm{det}} \otimes \mathcal{H}_{\mathrm{unobs}}$ decomposes into detected and unobserved particles. The *inclusive scattering channel* is

$$\Phi_S(\rho) \;=\; \mathrm{Tr}_{\mathrm{unobs}}(S\,\rho\,S^\dagger),$$

which is CPTP (since S is an isometry, Φ_S is a Stinespring dilation) with Kraus rank bounded by Choi's theorem: $r \leq d_{\mathrm{in}}$.

Conservation laws (total angular momentum J_z, parity P, charge, etc.) impose linear constraints on the Choi matrix J_{Φ_S}. If a conserved charge Q satisfies $[Q_{\mathrm{total}}, S] = 0$, then for the inclusive channel one derives constraints of the form $J_{(i,a),(j,b)} = 0$ whenever the corresponding quantum numbers are incompatible (e.g. $J_z^{(i)} + S_z^{(a)} \neq J_z^{(j)} + S_z^{(b)}$). These are linear constraints on the entries of J.

By Theorem 7, the total constraint count m (including both the TP condition and conservation law constraints) determines the BP bound:

$$r \;\leq\; \lfloor\sqrt{m}\,\rfloor \qquad \text{where } m = m_{\mathrm{TP}} + m_{\mathrm{cons}}.$$

Here m_{TP} is the number of independent real constraints from $\mathrm{Tr}_{\mathrm{det}}(J) = I$, and m_{cons} is the number of independent zero-entry constraints from conservation laws. Physically, r bounds the number of distinct scattering pathways that contribute to the detected particle's state.

13.5 Scattering example 1: generic $2 \to 2$ with spin

Consider two incoming particles with total input Hilbert space dimension d_{in}, detecting one outgoing particle in a d_{det}-dimensional space. With k conservation laws:

d_{in}	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 5$	$k = 10$
2	2	2	2	2	3	3
3	3	3	3	3	3	4
4	4	4	4	4	4	5
6	6	6	6	6	6	6
8	8	8	8	8	8	8
10	10	10	10	10	10	10

Values show $r_{\max} = \lfloor\sqrt{d_{\mathrm{in}}^2 + k}\rfloor$, treating $m_{\mathrm{TP}} = d_{\mathrm{in}}^2$ (complex Choi matrix) and k additional constraints from conservation laws. For moderate $k \ll d_{\mathrm{in}}^2$, the bound remains close to d_{in}. Conservation laws have significant impact only when k is comparable to d_{in}^2.

13.6 Scattering example 2: electron–electron (Coulomb)

For $e^- + e^- \to e^- + e^-$ (Coulomb scattering), we use a partial-wave decomposition truncated at orbital angular momentum $L \leq L_{\max}$. The basis states are $|L, M_L, S, M_S\rangle$ with $L + S$ even (antisymmetry of identical fermions), $S \in \{0, 1\}$ (singlet/triplet). The inclusive channel traces out one electron, detecting the other's spin ($d_{\text{det}} = 2$).

Conservation of $J_z = M_L + M_S$ and parity $P = (-1)^L$ impose linear constraints on the Choi matrix. The SDP verification (`numerics/scattering_bp.jl`) confirms the BP bound:

$L_{\max}$	d_{in}	d_{det}	m_{TP}	m_{J_z}	m_P	r_{BP}	r_{obs}
1	10	2	55	154	10	20	9
2	15	2	120	357	44	31	14
3	36	2	666	2216	114	76	–
4	45	2	1035	3515	232	97	–

The SDP uses a real symmetric Choi matrix (the partial-wave S-matrix is real by time-reversal symmetry), so $m_{\text{TP}} = d_{\text{in}}(d_{\text{in}} + 1)/2$ and the BP bound is $r(r + 1)/2 \leq m$ (the real symmetric formula from Theorem 6). For $L_{\max} \leq 2$, the SDP is solved numerically and the observed ranks confirm $r_{\text{obs}} \leq r_{\text{BP}}$. For $L_{\max} \geq 3$, the Choi matrix exceeds 70×70; dashes indicate analytical bounds only.

13.7 Scattering example 3: Compton scattering

For $\gamma + e^- \to \gamma + e^-$ (Compton scattering), the photon has helicities $\lambda_\gamma = \pm 1$ and the electron $\lambda_e = \pm\frac{1}{2}$. In the partial-wave (helicity) basis $|J, \lambda_\gamma, \lambda_e\rangle$ with half-integer J up to $J_{\max}$, the inclusive channel traces out the photon and observes the electron spin ($d_{\text{det}} = 2$).

Conservation of total $J_z = \lambda_\gamma + \lambda_e$ block-diagonalises the Choi matrix. The BP bound on the Kraus rank of the detected-electron channel is verified numerically:

$J_{\max}$	d_{in}	d_{det}	m_{TP}	m_{cons}	r_{BP}	r_{obs}
1	6	2	21	54	11	5
2	10	2	55	153	19	8
3	14	2	105	302	28	10
4	18	2	171	501	36	14

All four cases are solved numerically using a real symmetric SDP (with $r(r + 1)/2 \leq m$ as the BP formula and $m_{\text{TP}} = d_{\text{in}}(d_{\text{in}} + 1)/2$). As $J_{\max}$ increases, d_{in} grows linearly and the BP bound scales as $r \sim d_{\text{in}}$. The observed ranks are significantly below the BP ceiling: helicity conservation block-diagonalises the Choi matrix, reducing the effective number of free parameters.

14 BP for Quantum Error Correction

The Barvinok–Pataki theorem gives structural bounds on optimal recovery channels in quantum error correction (QEC). The Knill–Laflamme conditions [11] that characterise correctable codes are *linear* constraints on the recovery channel's Choi matrix, making the recovery problem a spectrahedron.

14.1 Knill–Laflamme conditions as linear constraints

Let $\mathcal{C} \subseteq \mathbb{C}^d$ be a k-dimensional code subspace with projector $P_\mathcal{C}$, and let $\{E_a\}_{a=1}^N$ be the Kraus operators of a noise channel $\mathcal{E}$. The code $\mathcal{C}$ can detect the errors $\{E_a\}$ if and only if

$$P_\mathcal{C}\, E_a^\dagger E_b\, P_\mathcal{C} \;=\; C_{ab}\, P_\mathcal{C} \qquad \forall\, a, b, \tag{3}$$

for some Hermitian matrix $C = (C_{ab})$ [11, 12]. In the orthonormal basis $\{|i_L\rangle\}_{i=1}^k$ of $\mathcal{C}$, this becomes $\langle i_L|E_a^\dagger E_b\,|j_L\rangle\rangle = C_{ab}\,\delta_{ij}$, which is a system of $N(N{+}1)/2 \cdot k(k{-}1)/2$ off-diagonal zero conditions and $N(N{+}1)/2{\cdot}(k{-}1)$ diagonal equality conditions—all *linear* in the entries of $E_a^\dagger E_b$.

14.2 BP bound for recovery channels

A recovery channel $\mathcal{R}\colon M_d \to M_d$ is represented by its Choi matrix $J_\mathcal{R} \in \mathrm{PSD}_{d^2}$ (real symmetric formulation), subject to:

1. **TP constraint:** $\mathrm{Tr}_{\mathrm{out}}(J_\mathcal{R}) = I_d$, giving $m_{\mathrm{TP}} = d(d{+}1)/2$ real constraints.
2. **KL constraints:** the correction condition $P_\mathcal{C}\,\mathcal{R}(\mathcal{E}(\rho))\,P_\mathcal{C} = \rho$ for all code states ρ translates to m_{KL} additional linear constraints on $J_\mathcal{R}$.

Corollary 33 (BP for QEC recovery channels)**.** *If $\mathcal{R}$ is an extreme recovery channel satisfying the TP and KL constraints, then its Kraus rank satisfies*

$$\frac{r(r+1)}{2} \;\le\; m_{\mathrm{TP}} + m_{\mathrm{KL}}.$$

Proof. All constraints are linear equalities on $J_\mathcal{R} \in \mathrm{PSD}_{d^2}$. Apply Theorem 7. □

The BP bound limits the *structural complexity* of optimal recovery: even for large codes, the recovery channel has bounded Kraus rank, determined by the number of independent KL conditions and the TP constraint.

14.3 Examples: constraint counts and rank bounds

The $[[4, 2, 2]]$ detection code. This code encodes $k = 4$ logical states into $n = 4$ physical qubits ($d = 16$). With distance $d_{\min} = 2$, it detects single-qubit errors. The error set consists of $N = 1 + 3 \cdot 4 = 13$ operators (identity plus weight-1 Paulis), giving $N(N{+}1)/2 = 91$ pairs. Each pair contributes $k(k{-}1)/2 + (k{-}1) = 6 + 3 = 9$ KL conditions, for a total of $m_{\mathrm{KL}} = 819$. With $m_{\mathrm{TP}} = 16 \cdot 17/2 = 136$, the BP bound gives $r(r{+}1)/2 \le 955$, hence $r \le 43$. The Choi matrix is 256×256; the SDP is tractable with Mosek.

The $[[5, 1, 3]]$ perfect code [12]. Here $n = 5$, $k = 2$, $d_{\min} = 3$ (corrects 1 error). The error set has $N = 16$ operators, giving $m_{\mathrm{KL}} = 272$ conditions. With $m_{\mathrm{TP}} = 32 \cdot 33/2 = 528$: $r(r{+}1)/2 \le 800$, hence $r \le 39$.

Remark (vacuity of the combined bound). For these codes, the combined BP bound $r_{\mathrm{BP}}(\mathrm{TP{+}KL})$ *exceeds* the TP-only bound $r \le d$ (which is Choi's theorem). This is an instance of the general phenomenon in Remark 30: adding constraints increases m and hence weakens the BP bound. The QEC application thus illustrates a *limitation* of the naive BP bound. To obtain tighter bounds, one should exploit the block structure of the recovery channel under the code's stabiliser symmetry, applying BP independently to each symmetry-reduced block.

14.4 Numerical verification

We verified the BP bound for the $[[4, 2, 2]]$ code by solving the recovery channel SDP using Mosek. The Choi matrix $J_\mathcal{R} \in \mathrm{PSD}_{256}$ is optimised subject to TP and code subspace preservation constraints, with random objectives to push to extreme points. Over 10 trials, the observed maximum Kraus rank satisfies the BP bound; see `numerics/qec_bp.jl`.

Table 8: BP bounds for QEC recovery channels. The Choi matrix $J_{\mathcal{R}}$ has dimension $d^2 \times d^2$ with $d = 2^n$. The BP bound on Kraus rank is $r \leq \lfloor(-1+\sqrt{1+8m})/2\rfloor$.

Code	n	k	$d_{\min}$	m_{TP}	m_{KL}	r_{BP}
$[[4,2,2]]$	4	4	2	136	819	43
$[[5,1,3]]$	5	2	3	528	272	39
$[[7,1,3]]$	7	2	3	8256	506	131

15 BP for the Quantum Marginal Problem

The *quantum marginal problem* asks: given reduced density matrices ρ_A and ρ_B, does there exist a joint state $\rho_{AB} \geq 0$ with $\mathrm{Tr}_B(\rho_{AB}) = \rho_A$ and $\mathrm{Tr}_A(\rho_{AB}) = \rho_B$? The set of compatible joint states is a spectrahedron, and the BP bound controls the rank of its extreme points.

15.1 Marginal constraints as a spectrahedron

Let $d_A, d_B \geq 2$ and $d = d_A d_B$. Fix Hermitian matrices $\rho_A \in \mathrm{PSD}_{d_A}$ and $\rho_B \in \mathrm{PSD}_{d_B}$ with $\mathrm{Tr}(\rho_A) = \mathrm{Tr}(\rho_B) = 1$. The set of compatible joint states is

$$\mathcal{M}(\rho_A, \rho_B) \;=\; \left\{ \rho_{AB} \in \mathrm{PSD}_d \;\middle|\; \mathrm{Tr}_B(\rho_{AB}) = \rho_A,\ \mathrm{Tr}_A(\rho_{AB}) = \rho_B \right\}.$$

The marginal constraints are affine equalities on ρ_{AB}: the partial traces are linear maps, and the target values ρ_A, ρ_B are fixed. In the real symmetric formulation, the constraint counts are:

- $\mathrm{Tr}_B(\rho_{AB}) = \rho_A$: $d_A(d_A{+}1)/2$ real conditions.
- $\mathrm{Tr}_A(\rho_{AB}) = \rho_B$: $d_B(d_B{+}1)/2$ real conditions.
- One constraint is redundant: $\mathrm{Tr}(\rho_A) = \mathrm{Tr}(\rho_B) = 1$ is implied by both marginals.

15.2 BP bound for marginal-constrained states

Corollary 34 (BP for quantum marginals)**.** *Let ρ_{AB} be an extreme point of $\mathcal{M}(\rho_A, \rho_B)$. Then*

$$\frac{r(r+1)}{2} \;\leq\; \frac{d_A(d_A{+}1)}{2} + \frac{d_B(d_B{+}1)}{2} - 1,$$

where $r = \mathrm{rank}(\rho_{AB})$.

Proof. The total number of independent real constraints is $m = d_A(d_A{+}1)/2 + d_B(d_B{+}1)/2 - 1$. Apply Theorem 7 to $\rho_{AB} \in \mathrm{PSD}_d$. □

For small systems, the bound is quite restrictive:

d_A	d_B	$d = d_A d_B$	m_A	m_B	m	r_{BP}
2	2	4	3	3	5	2
2	3	6	3	6	8	3
2	4	8	3	10	12	4
3	3	9	6	6	11	4
3	4	12	6	10	15	5
4	4	16	10	10	19	5

For two qubits ($d_A = d_B = 2$): $r \leq 2$, meaning extreme compatible states have rank at most 2. This is consistent with the known result that compatible two-qubit states with fixed marginals can always be decomposed into at most two pure states.

15.3 Connection to quantum de Finetti

The BP bound for the marginal problem complements the structural results of [13] on quantum marginal compatibility. While the marginal problem asks *which* reduced states are compatible, the BP bound controls the *rank* of extreme compatible states, bounding the entanglement complexity. This is also related to quantum de Finetti theorems: rather than bounding the distance to a separable state, BP bounds the rank of extremals, which controls the number of terms needed in a convex decomposition.

For a bipartite system with fixed marginals, the extreme compatible states have rank at most r_{BP}, which is independent of how the marginals were prepared. This means that the entanglement in any compatible state can be "concentrated" into at most r_{BP} Schmidt coefficients by moving to an extreme point.

15.4 Numerical verification

We verified the BP bound by solving the marginal-constrained SDP for qubit-qubit (2×2), qubit-qutrit (2×3), and qutrit-qutrit (3×3) systems, using Mosek with random objectives to generically select extreme points.

Table 9: Numerical verification of the BP bound for marginal-constrained states. All observed ranks satisfy $r(r+1)/2 \leq m$. The marginals are either maximally mixed, pure, or randomly generated.

d_A	d_B	**Marginals**	m	r_{obs}	r_{BP}	
2	2	max. mixed	5	1	2	✓
2	2	both pure	5	1	2	✓
2	2	random	5	2	2	✓
2	3	max. mixed	8	3	3	✓
2	3	random	8	3	3	✓
3	3	max. mixed	11	2	4	✓
3	3	random	11	3	4	✓

See `numerics/marginals_bp.jl` for full numerical details. For qubit-qubit, extreme compatible states generically have rank 2, saturating the BP bound. For larger systems, the bound is typically not saturated: the solver finds extreme points with rank below the ceiling, reflecting the geometry of the compatible set.

16 BP for Thermal Operations

Thermal operations—quantum channels that preserve the Gibbs state—form a spectrahedron to which the BP bound applies. This gives structural bounds on the Kraus complexity of extreme thermal operations, relevant for resource theories of thermodynamics [14, 15].

16.1 Thermal operations as a spectrahedron

Let H be a Hamiltonian on a d-dimensional system with eigenvalues $E_1 \leq \cdots \leq E_d$, and let $\gamma_\beta = e^{-\beta H}/Z$ be the Gibbs state at inverse temperature $\beta = 1/(k_B T)$. A *Gibbs-preserving channel* is a CPTP map $\Phi\colon M_d \to M_d$ satisfying $\Phi(\gamma_\beta) = \gamma_\beta$.

In the Choi picture, the Gibbs preservation condition becomes

$$\mathrm{Tr}_{\mathrm{in}}\big((\gamma_\beta^\top \otimes I)\, J_\Phi\big) \;=\; \gamma_\beta,$$

which is a system of $d(d{+}1)/2$ real-linear constraints on J_Φ (the upper triangle of the $d \times d$ Hermitian target γ_β). Combined with the TP constraint $\mathrm{Tr}_{\mathrm{out}}(J_\Phi) = I_d$ ($d(d{+}1)/2$ constraints), the total is

$$m \;=\; d(d{+}1)/2 + d(d{+}1)/2 - 1 \;=\; d(d{+}1) - 1,$$

where one constraint is redundant ($\mathrm{Tr}(\Phi(\gamma_\beta)) = 1$ is implied by TP).

16.2 BP bound for extreme thermal operations

Corollary 35 (BP for thermal operations)**.** *Let Φ be an extreme Gibbs-preserving CPTP channel on M_d. Then the Kraus rank satisfies*

$$\frac{r(r+1)}{2} \;\leq\; d(d{+}1) - 1.$$

Proof. Apply Theorem 7 to $J_\Phi \in \mathrm{PSD}_{d^2}$ with $m = d(d{+}1) - 1$ constraints. □

For small systems:

d	m_{TP}	m_{Gibbs}	m	r_{BP}	r_{obs} (β=1)	
2	3	3	5	2	2	✓
3	6	6	11	4	3	✓
4	10	10	19	5	4	✓
5	15	15	29	7	4	✓

16.3 Temperature dependence

The constraint count $m = d(d{+}1) - 1$ is independent of β, so the BP bound on Kraus rank is temperature-independent. However, the *effective* rank of extreme thermal operations depends on temperature through two limiting cases:

- **High temperature ($\beta \to 0$):** $\gamma_\beta \to I/d$, so Gibbs preservation becomes *unitality* ($\Phi(I/d) = I/d$, i.e. doubly stochastic). For qubits at $\beta = 0$, extremal doubly stochastic channels have rank 1 (unitary channels), below the BP bound $r \leq 2$.
- **Low temperature ($\beta \to \infty$):** $\gamma_\beta \to |0\rangle\langle 0|$, so Gibbs preservation becomes $\Phi(|0\rangle\langle 0|) = |0\rangle\langle 0|$ (ground-state preservation). The constraint count $m = d(d{+}1) - 1$ remains the same, but the Gibbs constraints become highly degenerate (rank-1 target). The observed extremal ranks generically saturate the BP bound.

16.4 Numerical verification

We verified the BP bound for qubits ($d = 2$) and qutrits ($d = 3$) at inverse temperatures $\beta \in \{0, 0.5, 1, 2, 5\}$ by solving the Gibbs-preserving channel SDP with Mosek. For qubits at $\beta = 0$, the extremal rank is 1 (unitary channels); for $\beta > 0$, the rank is generically 2, saturating the bound. For qutrits at $\beta = 0.5$, the rank reaches 4, saturating $r_{\mathrm{BP}} = 4$. See `numerics/thermal_bp.jl` for full data.

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Archimedean order unit spaces. Martin Plavala recognised during the discussion that the argument extends to general cones. Alexander Stottmeister pointed out that the face-dimension quantity d_F can be controlled in infinite dimensions via Theorem 3.3 of Weis–Shirokov [4], which requires only a real vector space and a convex set, without operator-algebraic assumptions. The recording was transcribed and typeset with the assistance of Claude (Anthropic).

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